

KEYSTONE LIBRARY



CONNOR LANGSFORD

Keystone Library

Connor Langsford

2/21/25

Abstract

Design documents of the creation of a Library in Noblesville, IN

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Project Description

-Overview

Design of the redevelopment of an existing industrial brownfield site in Noblesville, IN.

-Criteria and Constraints

- A safe and efficient building with adequate utilities and services
- parking with desirable traffic flow
- storm water management
- modern entryway and vestibule
- accessibility
- exterior covered area with benches
- Elevator must be added
- One Low Impact Development technique
- Entry and Circulation Desk area
- Children's Help Desk
- Open area for stacks and computer terminals on both floors
- Public restroom on both floors
- Head Librarian's office
- Staff workroom/ kitchenette
- Staff restroom
- Two meeting rooms
- Noblesville History room
- Mechanical room
- Books out of direct sunlight
- Green roof with walkway
- Trash receptacle

Building Code and Regulations

-Building Type

This library is public so it will be a type one construction using non - combustible materials. (Reference IBC, chapters 5-6)

-Occupancy

Group A – 3 assembly (Reference IBC, chapter 3)

-Egress

Must meet Egress requirements for an A – 3 building (Reference IBC, chapter 10)

- Number of exits: 2 per story (Reference IBC, Table 1021.1)
- Width of exits: 0.3in stairway per occupant, 0.2in door per occupant
(Reference IBC, Section 3412.6.11)

$$\text{Stairway width} = 0.3in \times 194 \text{ people}$$

$$\text{Stairway width} = 58.2in$$

$$\text{Door width} = 0.2in \times 194 \text{ people}$$

$$\text{Door width} = 38.8in$$

-Climate Zone

Zone 5A, cold and moist (Reference Energy Conservation Code)

-Zoning

B: Commercial District (Reference NUDO, Article 7)

- Minimum Lot Size

3 acers (Reference FUDO)

-Maximum Building Hight

35ft (Reference FUDO)

-Roof Live Load

20psf but will be adjusted based on snow load calculations (Reference IBC, chapter 16, section 1607)

-Parking Lot

- Ninety-degree (90°) angle space: Twenty-four-foot (24') wide parking aisle for one- or two-way traffic, 3ft wide one-way sidewalk, 5ft wide two-way sidewalk (Reference, FUDO)
- Driveway Width must be at least 10ft but no more than 24ft wide.
- Lights shall not exceed 35ft tall (Reference, FUDO), constant throughout property
- One space per 3 people (Reference FUDO)

$$\text{Reading Area} = 50\text{sqft} * \text{person}$$

$$\text{Stack Area} = 100\text{sqft} * \text{person}$$

$$\text{Meeting Rooms} = 15\text{sqft} * \text{person}$$

$$\text{Office Space} = 150\text{sqft} * \text{person}$$

$$\text{Max Occupancy} = \frac{\text{Library Area (sqft)}}{\text{Occupant Load Factor (sqft * person)}}$$

$$\text{Reading Area} = \frac{6120\text{sqft}}{50\text{sqft} * \text{person}}$$

$$\text{Reading Area} = 122 \text{ people}$$

$$\text{Stack Area} = \frac{4320\text{sqft}}{100\text{sqft} * \text{person}}$$

$$\text{Stack Area} = 43 \text{ people}$$

$$\text{Meeting Rooms} = \frac{360\text{sqft}}{15\text{sqft} * \text{person}}$$

$$\text{Meeting Rooms} = 24 \text{ people}$$

$$\text{Office Space} = \frac{720\text{sqft}}{150\text{sqft} * \text{person}}$$

$$\text{Office Space} = 5 \text{ people}$$

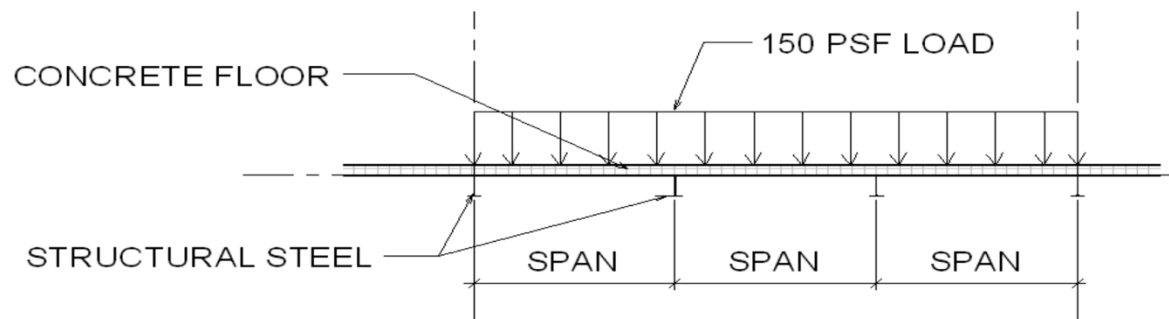
$$\text{Total Max Occupancy} = 194 \text{ people}$$

$$\text{Number of Parking Spaces} = \frac{\text{Max Occupancy}}{3}$$

$$\text{Number of Parking Spaces} = 65 \text{ spaces}$$

- 51 to 75 spaces need at least 3 accessible parking spaces (Reference IBC, Table 1106.1)
- Loading berths and areas shall be provided
- Screening shall be a ten-foot (10') wide landscaped buffer with One evergreen tree planted every eight (8) feet; and one shrub planted every twelve (12) feet; and one deciduous tree planted every fifteen (15) feet along the perimeter of the outdoor sales or outdoor display area (Reference FUDO)
- Each island or peninsula shall be at least one hundred eighty (180) square feet in size. 3. Each island or peninsula shall contain at least one (1) tree per one hundred eighty (180) square feet. (Reference FUDO)
- Minimum thirty-foot (30') buffer yard (Reference FUDO)
- Designed such that there will be no increase in the peak discharge runoff rate as a result of the proposed development (Reference FUDO)
- 10 to 99 spaces need 5% to be landscaped (Reference FUDO)
- Set back of 8ft from all buildings (Reference FUDO)

Composite Floor Load-Span table



Roof Load

-Live Load

20psf (Reference IBC 1607.11)

-Dead Load

Steel Joist 10psf, Plaster ceiling 10psf

-Snow Load

$$P_s = 0.7C_sC_eC_tI_sP_g$$

$$P_s = 0.7(1)(1)(1.1)(1.1)(20psf)$$

$$P_s = 16.94psf$$

$$P_s \geq I_sP_g$$

$$16.94 \geq (1.1)(20psf)$$

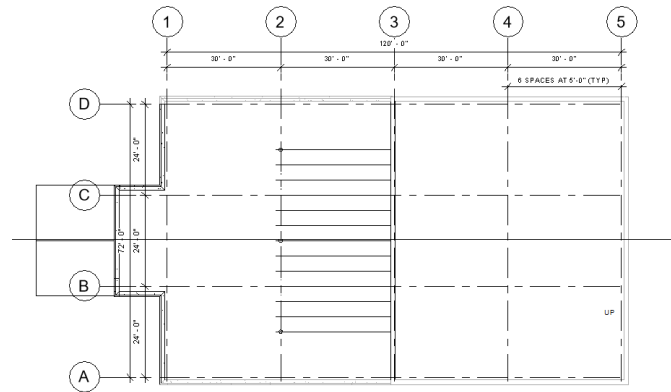
$$16.94 \geq 22sf$$

$$P_s = 22psf$$

-Total Roof Design Load

$$62psf$$

Beam Calculations



Dead Load = Roofing 62psf + mechanicals 3psf

(Reference IBS section 1606)

Live Load = stack room 150^bpsf (Reference IBS section 1607.1)

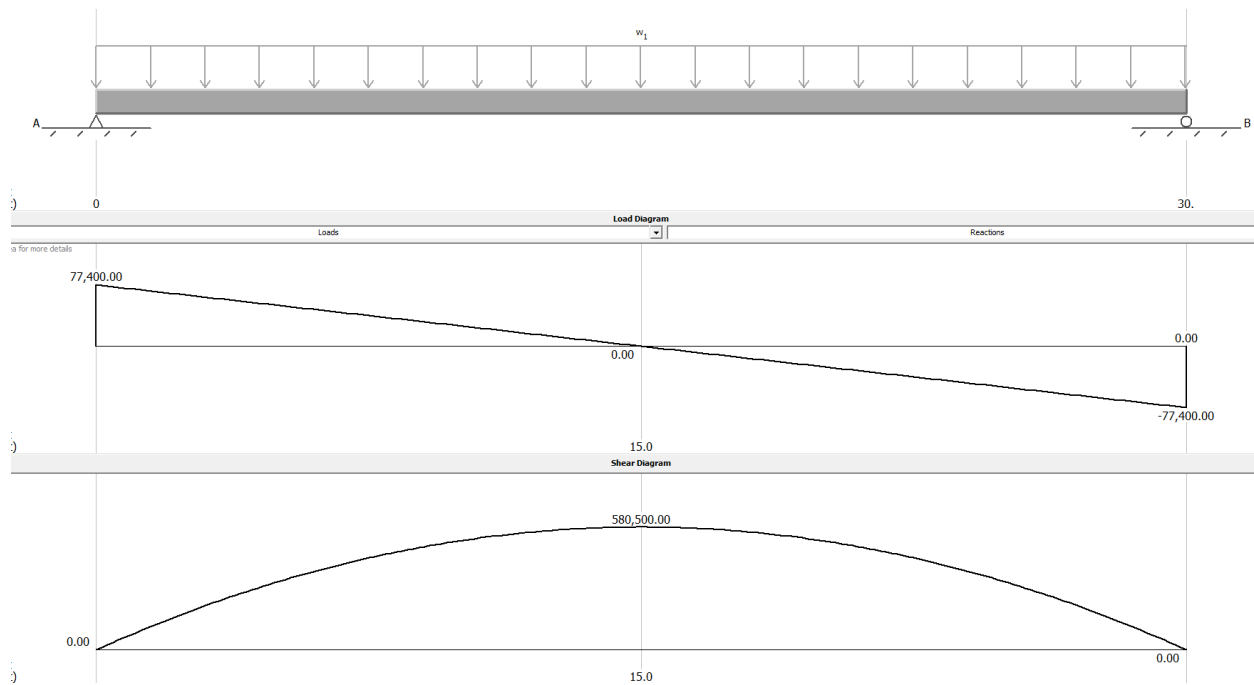
Total Load = 215psf

$F_y = 50 \text{ ksi}$

$E = 29,000 \text{ ksi}$

-Interior Beam

- Diagram



- Calculation

$$w = \text{Total loads psf} \times \text{Tributary Width ft}$$

$$w = 215\text{psf} \times 24\text{ft}$$

$$w = 5160\text{plf}$$

$$R_{a/b} = \frac{wL}{2}$$

$$R_{a/b} = \frac{(5160\text{plf})(30\text{ft})}{2}$$

$$R_{a/b} = 77400\text{lb}$$

$$M_{max} = \frac{wL^2}{8}$$

$$M_{max} = \frac{(5160\text{lb})(30\text{ft})^2}{8}$$

$$M_{max} = 580500\text{lb}\cdot\text{ft}$$

$$M_n \geq M_a \times \Omega_b$$

$$M_n \geq (580500lb * ft)(1.67)$$

$$M_n \geq 969435lb * ft$$

$$M_n = F_y \times Z_x$$

$$Z_x = \frac{M_n}{F_y}$$

$$Z_x = \frac{969435lb * ft}{50000psi} \times \frac{12ft}{1in}$$

$$Z_x = 232.66in^3$$

$$w27X84$$

$$V_n \geq V_a \times \Omega_v$$

$$V_n \geq (77400lb)(1.5)$$

$$V_n \geq 116100lb$$

$$V_n = 0.6 \times F_y \times d \times t_w$$

$$V_n = 0.6(50000psi)(26.71in)(0.46in)$$

$$V_n = 368598lb$$

$$368598lb \geq 116100lb$$

$$\Delta_{max} = \frac{5wL^4}{384EI}$$

$$\Delta_{max} = \frac{5(5160plf \times \frac{1in}{12ft})(30ft \times \frac{12in}{1ft})^4}{384(29000000psi)(2850in^4)}$$

$$\Delta_{max} = 1.14in$$

$$\frac{L}{240} > \Delta_{max}$$

$$\frac{(30ft \times \frac{12in}{1ft})}{240} > \Delta_{max}$$

$$1.5in > 1.14in$$

$$\frac{L}{360} > \Delta_{max}$$

$$\frac{(30ft \times \frac{12in}{1ft})}{360} > \Delta_{max}$$

$$1in > 1.14in$$

$$w27x94$$

$$\Delta_{max} = \frac{5(5160plf \times \frac{1in}{12ft})(30ft \times \frac{12in}{1ft})^4}{384(29000000psi)(3270in^4)}$$

$$\Delta_{max} = 0.99in$$

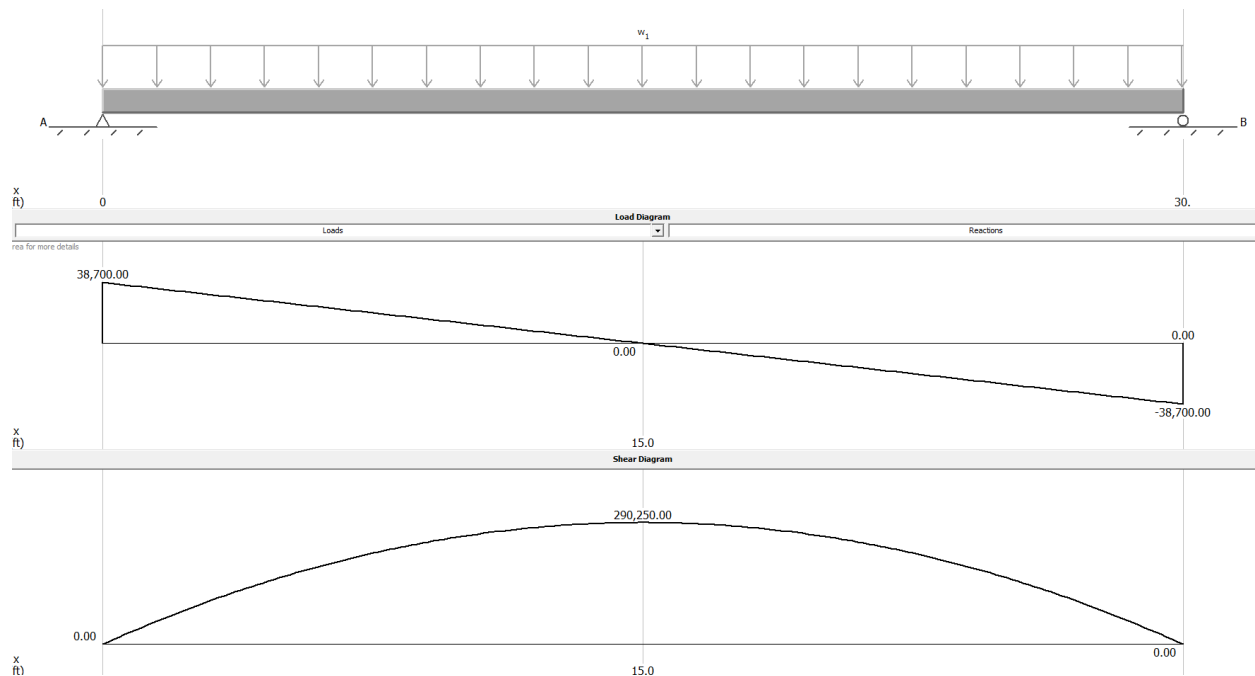
$$1 > 0.99in$$

Final Design: w30x90 because it is lighter than w27x94 and stiffer, so it still

meets all the requirements

-Exterior Beam

- Diagram



- Calculations

$$w = \text{Total loads psf} \times \text{Tributary Width ft}$$

$$w = 215\text{psf} \times 12\text{ft}$$

$$w = 2580\text{plf}$$

$$R_{a/b} = \frac{wL}{2}$$

$$R_{a/b} = \frac{(2580\text{plf})(30\text{ft})}{2}$$

$$R_{a/b} = 38700\text{lb}$$

$$M_{max} = \frac{wL^2}{8}$$

$$M_{max} = \frac{(2580\text{lb})(30\text{ft})^2}{8}$$

$$M_{max} = 290250\text{lb}\cdot\text{ft}$$

$$M_n \geq M_a \times \Omega_b$$

$$M_n \geq (290250lb * ft)(1.67)$$

$$M_n \geq 484717.5lb * ft$$

$$M_n = F_y \times Z_x$$

$$Z_x = \frac{M_n}{F_y}$$

$$Z_x = \frac{484717.5lb * ft}{50000psi} \times \frac{12ft}{1in}$$

$$Z_x = 116.33in^3$$

$$w24x55$$

$$V_n \geq V_a \times \Omega_v$$

$$V_n \geq (38700lb)(1.5)$$

$$V_n \geq 58050lb$$

$$V_n = 0.6 \times F_y \times d \times t_w$$

$$V_n = 0.6(50000psi)(23.57in)(0.395in)$$

$$V_n = 279304.5lb$$

$$279304.5lb \geq 58050lb$$

$$\Delta_{max} = \frac{5wL^4}{384EI}$$

$$\Delta_{max} = \frac{5(2580plf \times \frac{1in}{12ft})(30ft \times \frac{12in}{1ft})^4}{384(29000000psi)(1350in^4)}$$

$$\Delta_{max} = 1.20in$$

$$\frac{L}{240} > \Delta_{max}$$

$$\frac{(30ft \times \frac{12in}{1ft})}{240} > \Delta_{max}$$

$$1.5in > 1.2in$$

$$\frac{L}{360} > \Delta_{max}$$

$$\frac{(30ft \times \frac{12in}{1ft})}{360} > \Delta_{max}$$

$$1in > 1.2in$$

w24x68

$$\Delta_{max} = \frac{5(2580plf \times \frac{1in}{12ft})(30ft \times \frac{12in}{1ft})^4}{384(29000000psi)(1830in^4)}$$

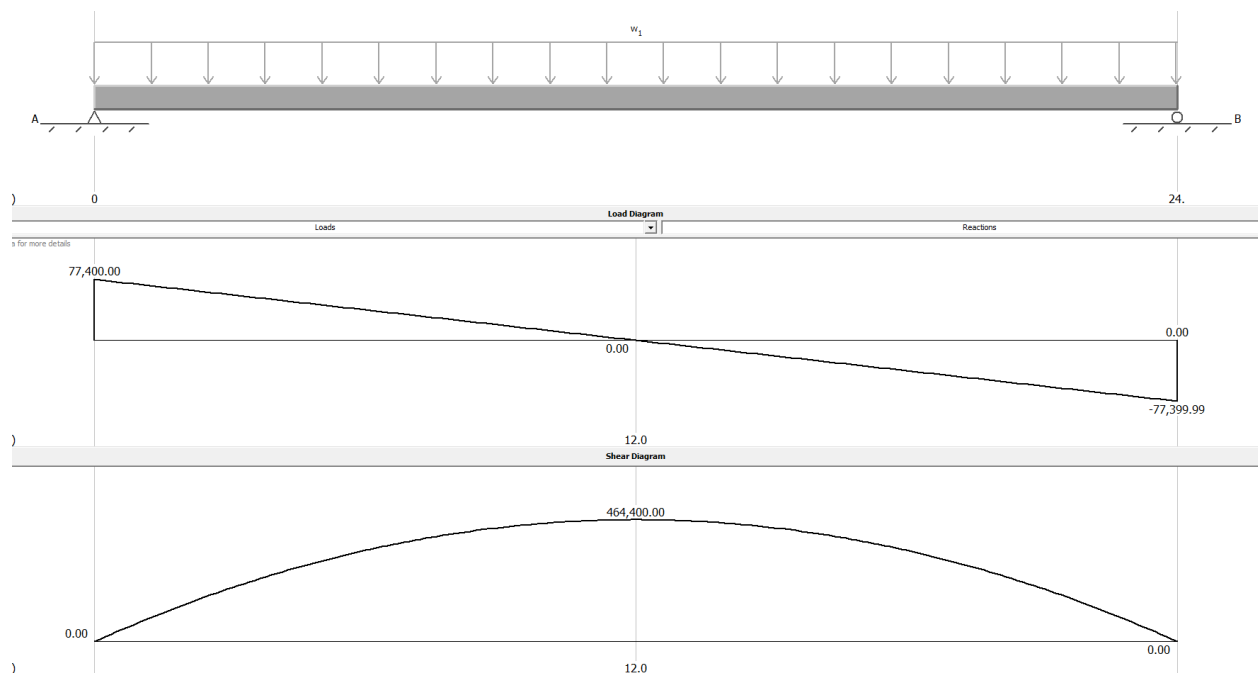
$$\Delta_{max} = 0.89in$$

$$1in > 0.89in$$

Final Design: w24x68

-Girder on Column line 3

- Diagram



- Calculation

$$w = \text{Total loads psf} \times \text{Tributary Width ft}$$

$$w = 215\text{psf} \times 30\text{ft}$$

$$w = 6450\text{plf}$$

$$R_{a/b} = \frac{wL}{2}$$

$$R_{a/b} = \frac{(6450\text{plf})(24\text{ft})}{2}$$

$$R_{a/b} = 77400\text{lb}$$

$$M_{max} = \frac{wL^2}{8}$$

$$M_{max} = \frac{(6450\text{lb})(24\text{ft})^2}{8}$$

$$M_{max} = 464400\text{lb}\cdot\text{ft}$$

$$M_n \geq M_a \times \Omega_b$$

$$M_n \geq (464400\text{lb} \cdot \text{ft})(1.67)$$

$$M_n \geq 775548\text{lb} \cdot \text{ft}$$

$$M_n = F_y \times Z_x$$

$$Z_x = \frac{M_n}{F_y}$$

$$Z_x = \frac{775548\text{lb} \cdot \text{ft}}{50000\text{psi}} \times \frac{12\text{ft}}{1\text{in}}$$

$$Z_x = 186.13\text{in}^3$$

$$\text{W27x84}$$

$$V_n \geq V_a \times \Omega_v$$

$$V_n \geq (77400lb)(1.5)$$

$$V_n \geq 116100lb$$

$$V_n = 0.6 \times F_y \times d \times t_w$$

$$V_n = 0.6(50000psi)(26.71in)(0.46in)$$

$$V_n = 368598lb$$

$$368598lb \geq 116100 lb$$

$$\Delta_{max} = \frac{5wL^4}{384EI}$$

$$\Delta_{max} = \frac{5(6450plf \times \frac{1in}{12ft})(24ft \times \frac{12in}{1ft})^4}{384(29000000psi)(2850in^4)}$$

$$\Delta_{max} = 0.58in$$

$$\frac{L}{240} > \Delta_{max}$$

$$\frac{(24ft \times \frac{12in}{1ft})}{240} > \Delta_{max}$$

$$1.2in > 0.58in$$

$$\frac{L}{360} > \Delta_{max}$$

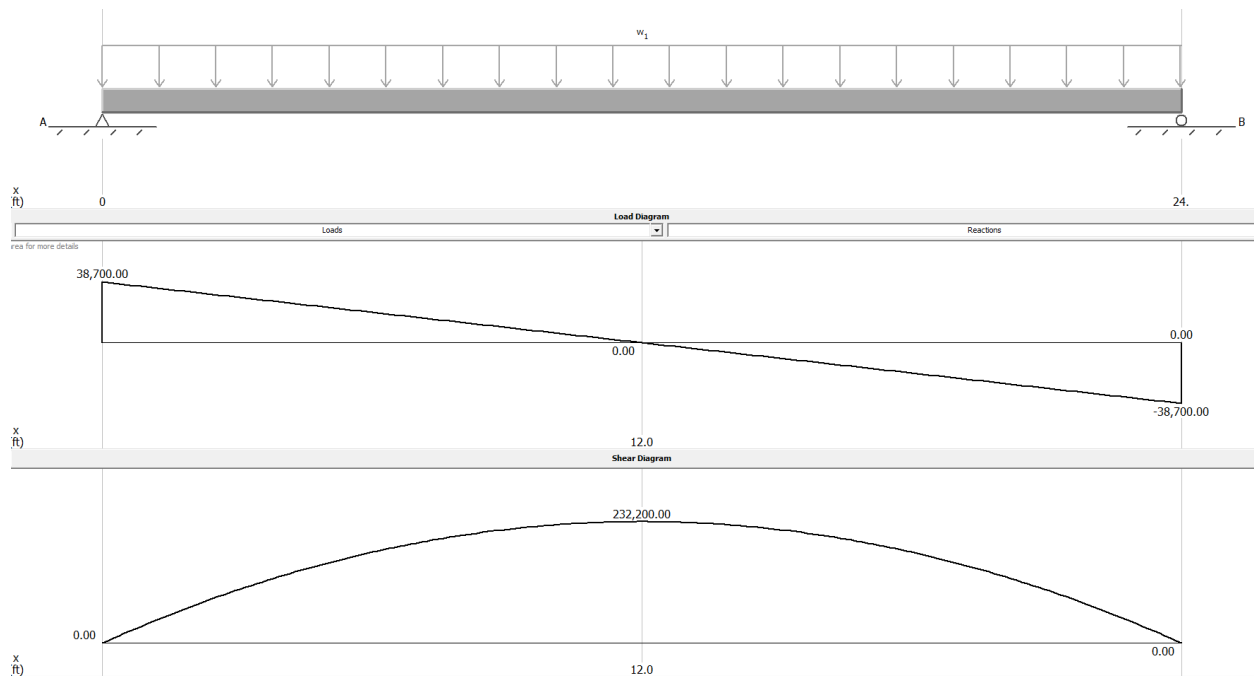
$$\frac{(24ft \times \frac{12in}{1ft})}{360} > \Delta_{max}$$

$$0.8in > 0.58in$$

Final Design: w27x84

-Girder on Column Line 5

- Diagram



- Calculation

$$w = \text{Total loads psf} \times \text{Tributary Width ft}$$

$$w = 215\text{psf} \times 15\text{ft}$$

$$w = 3225\text{plf}$$

$$R_{a/b} = \frac{wL}{2}$$

$$R_{a/b} = \frac{(3225\text{plf})(24\text{ft})}{2}$$

$$R_{a/b} = 38700\text{lb}$$

$$M_{max} = \frac{wL^2}{8}$$

$$M_{max} = \frac{(3225\text{lb})(24\text{ft})^2}{8}$$

$$M_{max} = 232200\text{lb}\cdot\text{ft}$$

$$M_n \geq M_a \times \Omega_b$$

$$M_n \geq (232200lb * ft)(1.67)$$

$$M_n \geq 387774lb * ft$$

$$M_n = F_y \times Z_x$$

$$Z_x = \frac{M_n}{F_y}$$

$$Z_x = \frac{387774lb * ft}{50000psi} \times \frac{12ft}{1in}$$

$$Z_x = 93.07in^3$$

$$W24x55$$

$$V_n \geq V_a \times \Omega_v$$

$$V_n \geq (38700lb)(1.5)$$

$$V_n \geq 58050lb$$

$$V_n = 0.6 \times F_y \times d \times t_w$$

$$V_n = 0.6(50000psi)(23.57in)(0.395in)$$

$$V_n = 279304.5lb$$

$$279304.5lb \geq 58050lb$$

$$\Delta_{max} = \frac{5wL^4}{384EI}$$

$$\Delta_{max} = \frac{5(3225plf \times \frac{1in}{12ft})(24ft \times \frac{12in}{1ft})^4}{384(29000000psi)(1350in^4)}$$

$$\Delta_{max} = 0.61in$$

$$\frac{L}{240} > \Delta_{max}$$

$$\frac{(24ft \times \frac{12in}{1ft})}{240} > \Delta_{max}$$

$$1.2in > 0.61in$$

$$\frac{L}{360} > \Delta_{max}$$

$$\frac{(24ft \times \frac{12in}{1ft})}{360} > \Delta_{max}$$

$$0.8in > 0.61in$$

Final Design: w24x55

Spread Footing

$$q_{allowable} = 3500psf$$

Current Footing: 7' x 7' x 2.03'

-Columns on Line 2

Interior Beam End Reaction = 77400lb

Interior Girder End Reaction = 77400lb

Total load = 309600lb

$$A \geq \frac{P}{q_{allowable}}$$

$$A \geq \frac{309600lb}{3500psf}$$

$$A \geq 88.46sf$$

$$A = 99.47sf$$

Footing: 7' x 7' x 2.03'

-Columns on Line 3

Green Roof = 309600lb

$$Roof = 62psf \times 180sf$$

$$Roof = 11160lb$$

Total load = 320760lb

$$A \geq \frac{P}{q_{allowable}}$$

$$A \geq \frac{320760lb}{3500psf}$$

$$A \geq 91.65sf$$

$$A = 99.47sf$$

Footing: 7' x 7' x 2.03'

-Columns on Line 4

Green Roof = 309600lb

$Roof = 62psf \times 720sf$

$Roof = 44640lb$

Total load = 354240lb

$$A \geq \frac{P}{q_{allowable}}$$

$$A \geq \frac{354240lb}{3500psf}$$

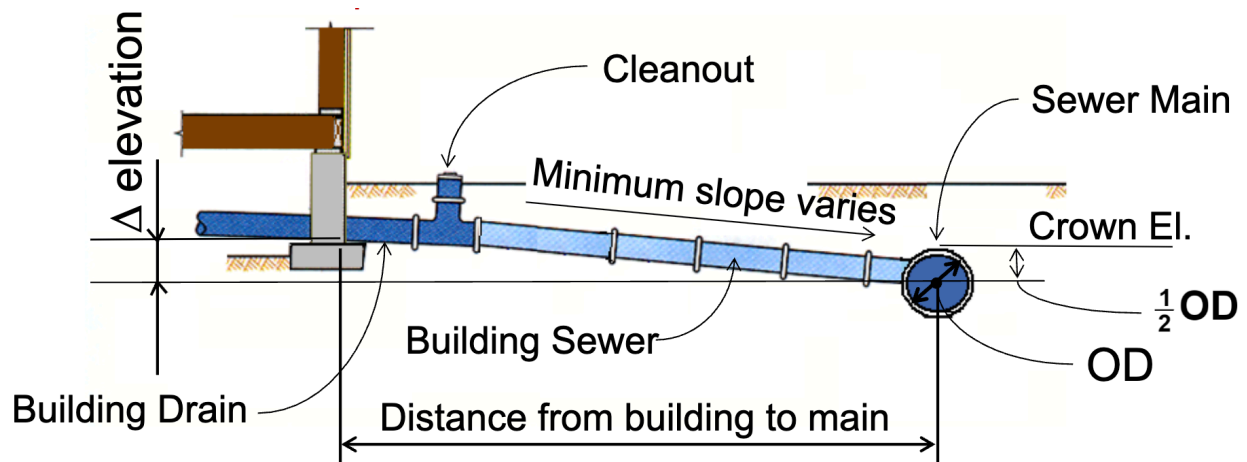
$$A \geq 101.21sf$$

$$A = 99.47sf$$

New footing = 8' x 8' x 2.03'

Wastewater Discharge

- Invert elevation at sewer main is 766.05ft
- 12in Pipe Diameter (1ft)



-Crown Elevation

$$\text{Crown Elevation} = \text{Invert Elevation} + \text{Pipe Diameter}$$

$$\text{Crown Elevation} = 766.05\text{ft} + 1\text{ft}$$

$$\text{Crown Elevation} = 767.05\text{ft}$$

-Distance from building to main = 123ft

-Slope

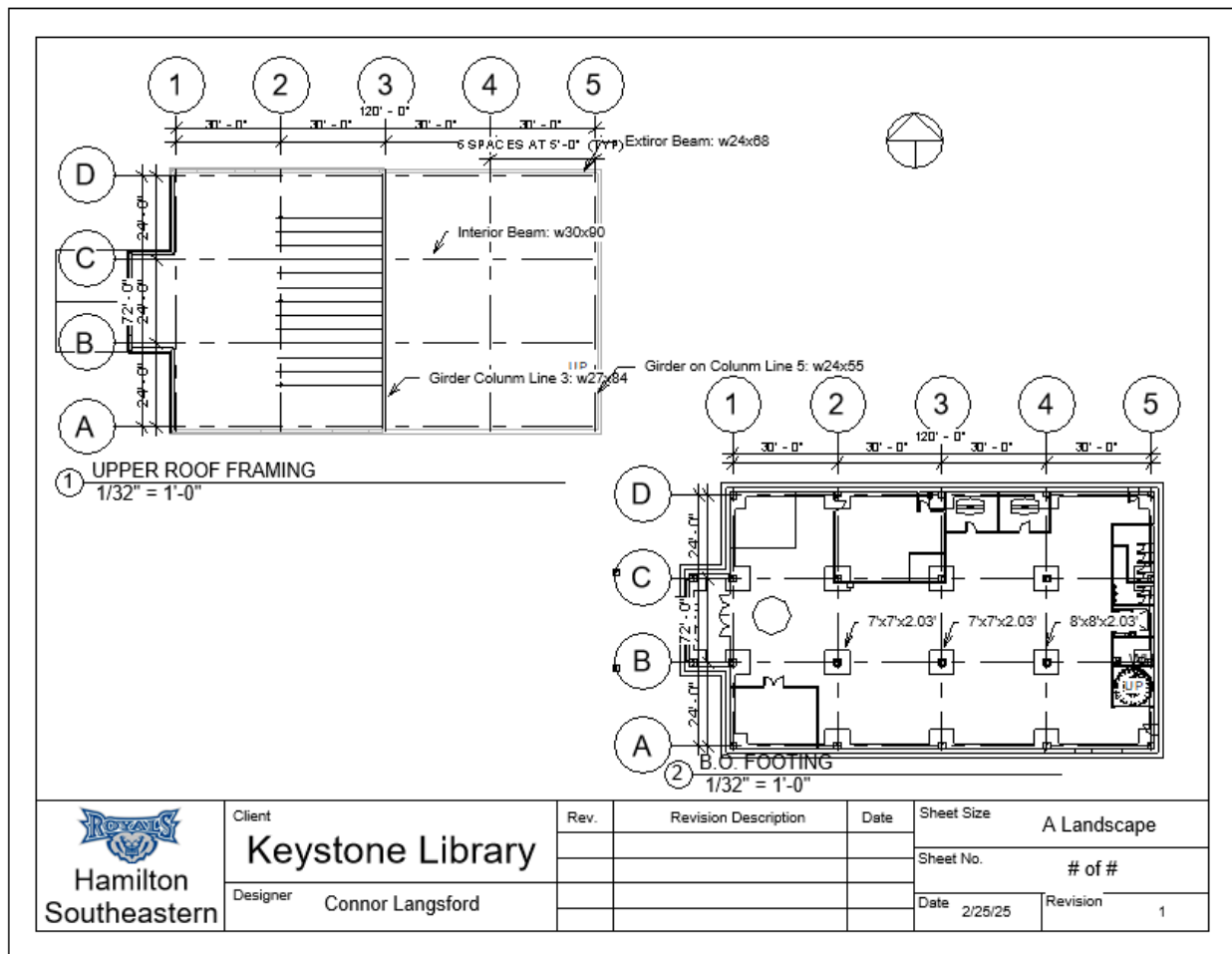
$$\text{Sewer Lateral Slope} = \frac{\text{inv. El building} - \text{Crown EL} + \frac{1}{2} \text{OD}}{\text{Distance}}$$

$$\text{Slope} = \frac{778\text{ft} - 767.05\text{ft} + (.5 \times .5\text{ft})}{416\text{ft}} * 100$$

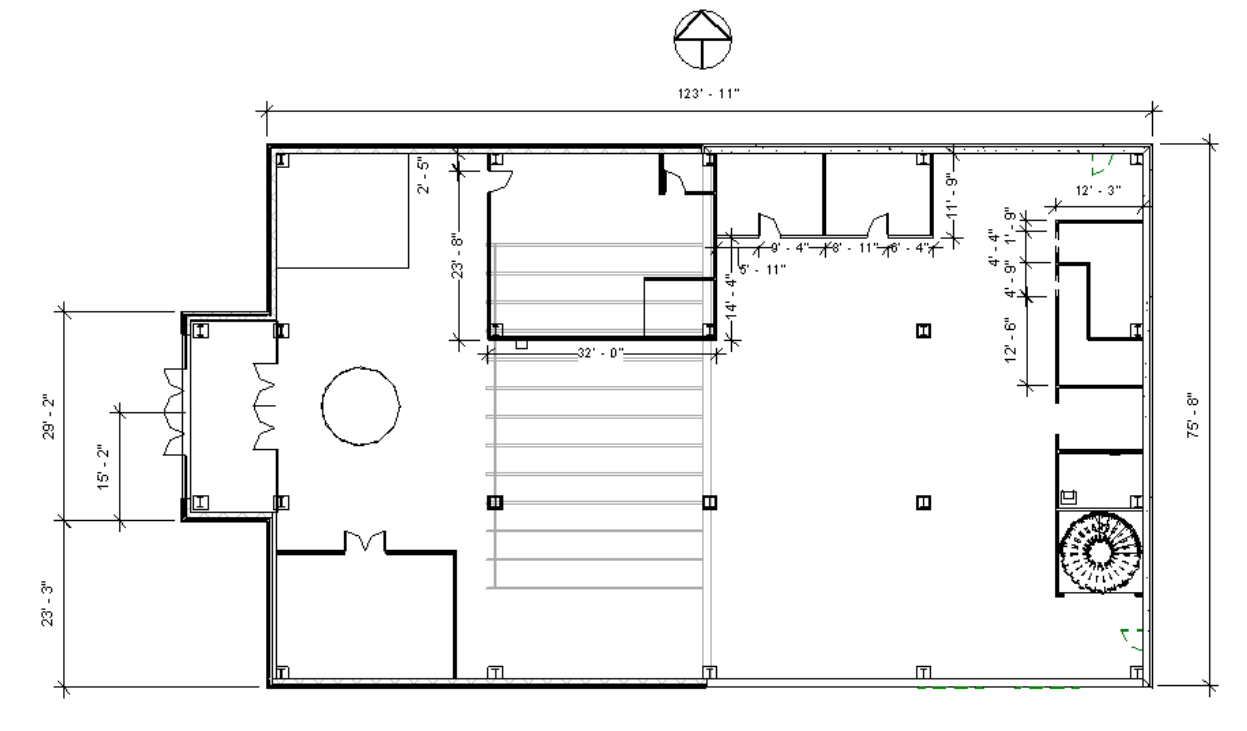
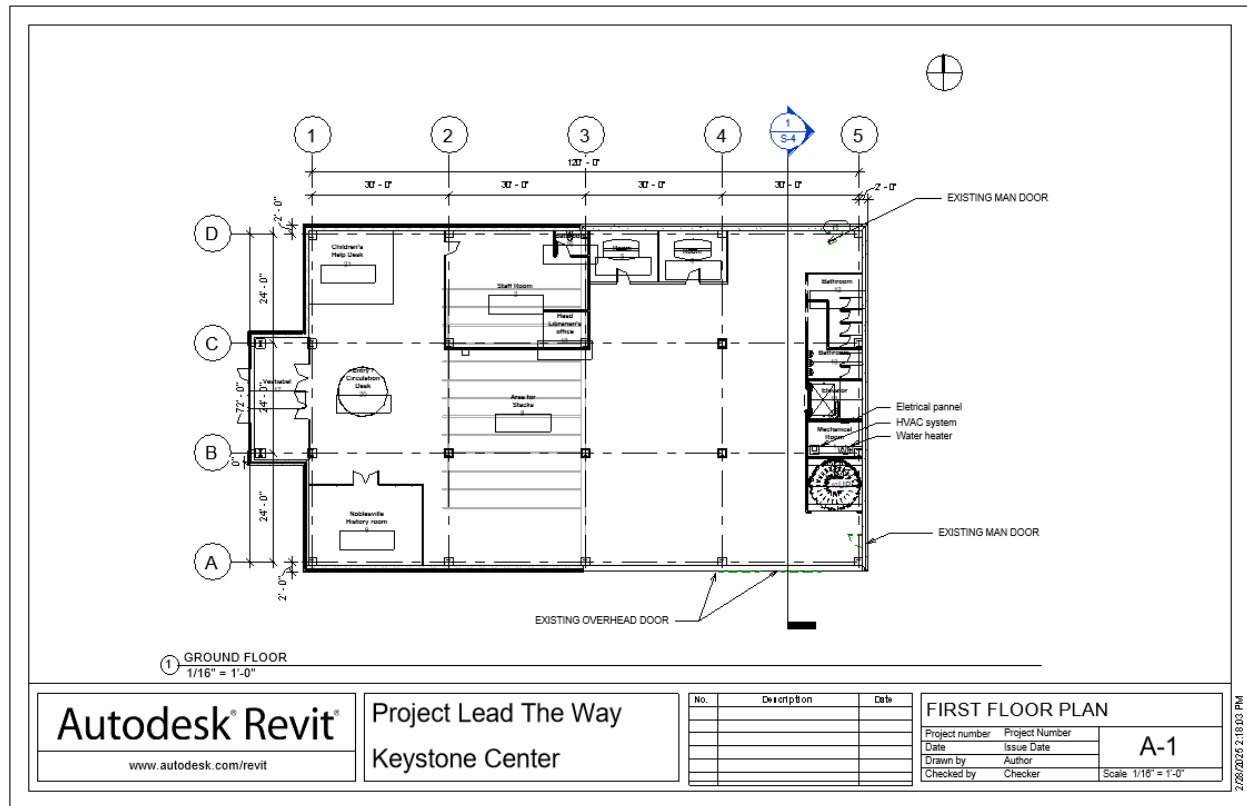
$$\text{Slope} = 2.6\%$$

$$\text{min slope} = 0.2\% \text{ (Reference IPC)}$$

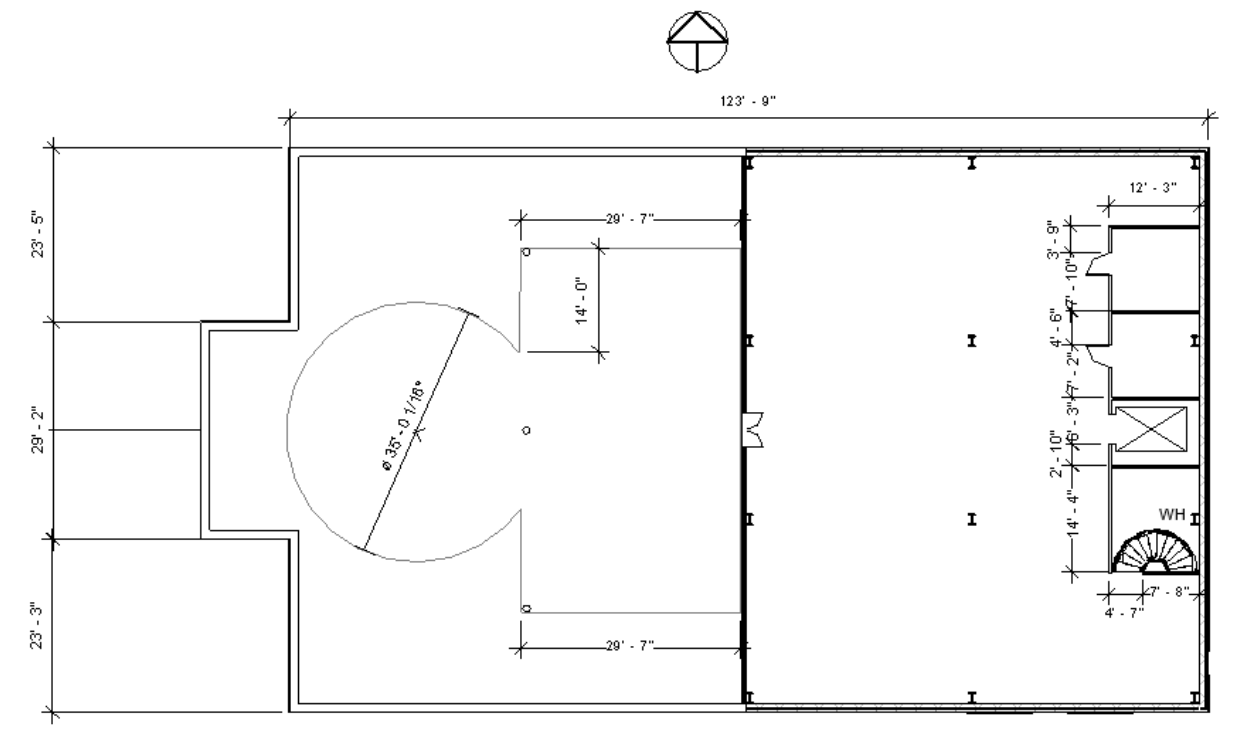
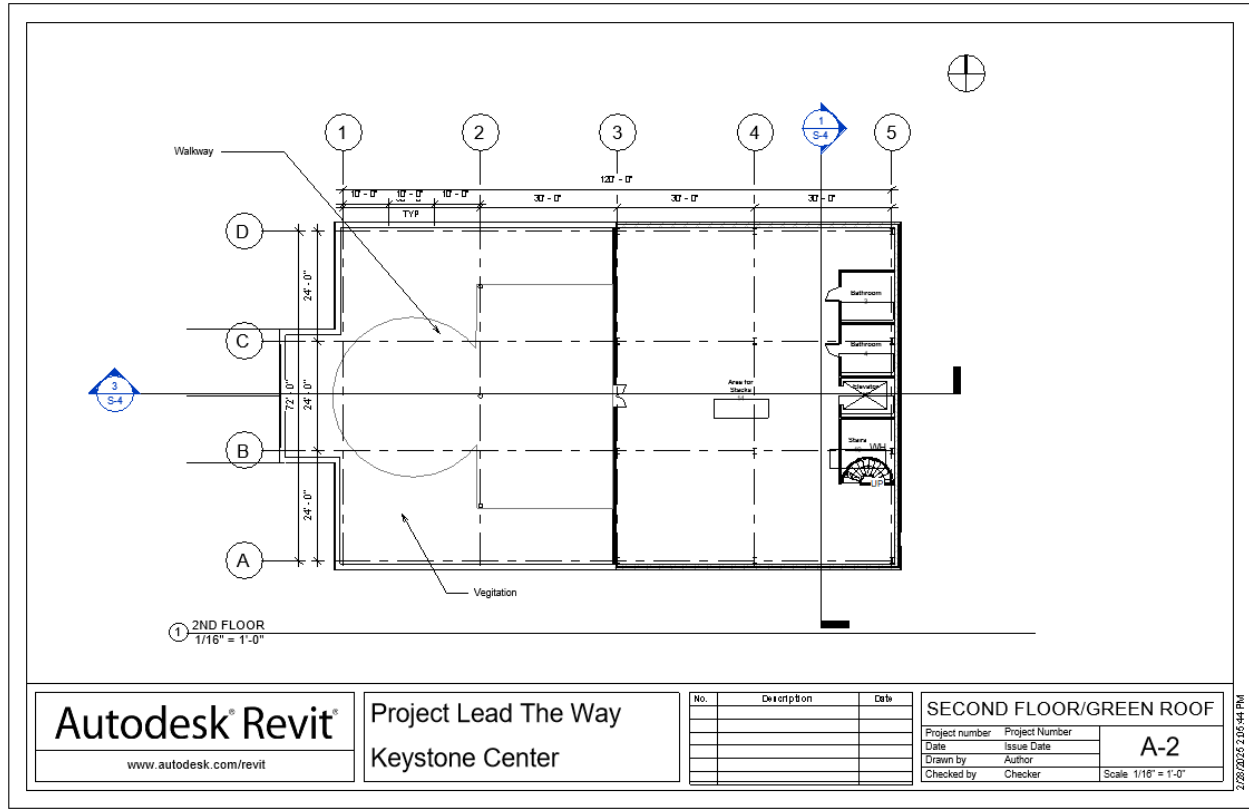
Framing



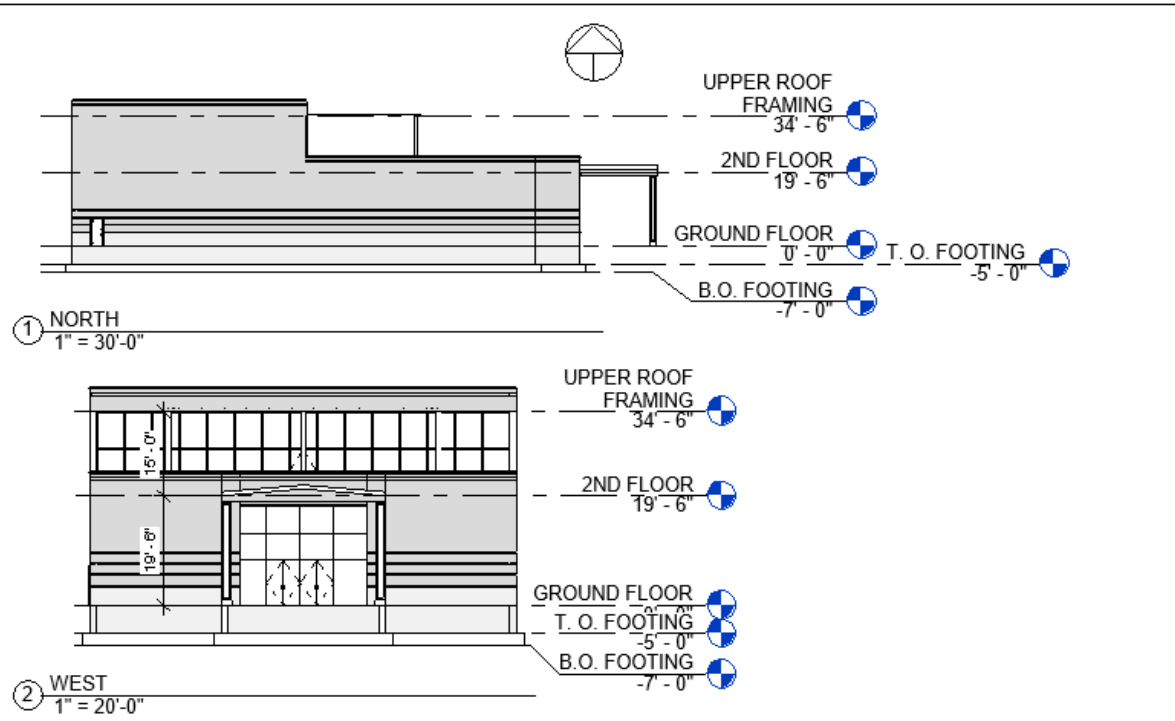
First Floor Plan



Second Floor Plan

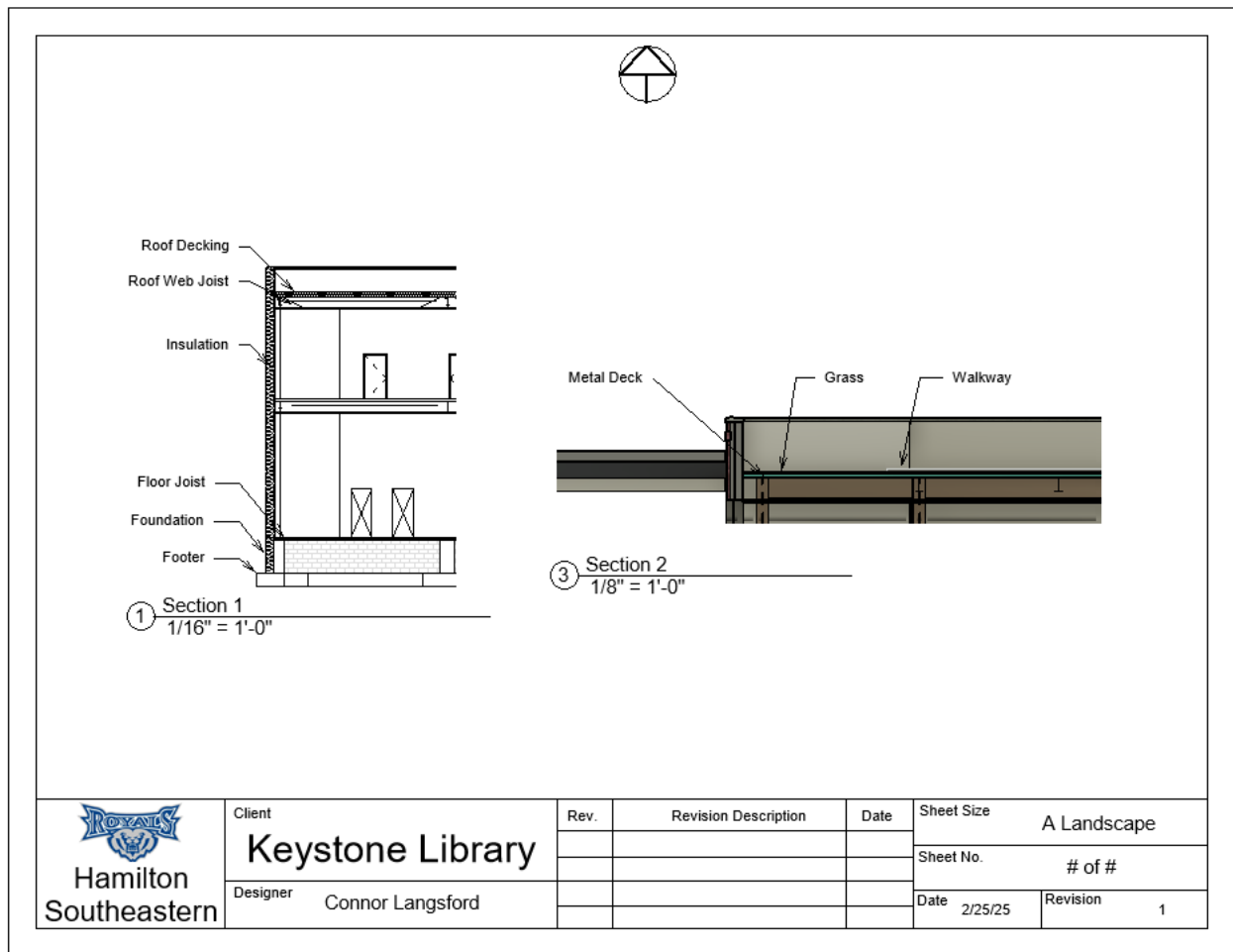


Elevation Views



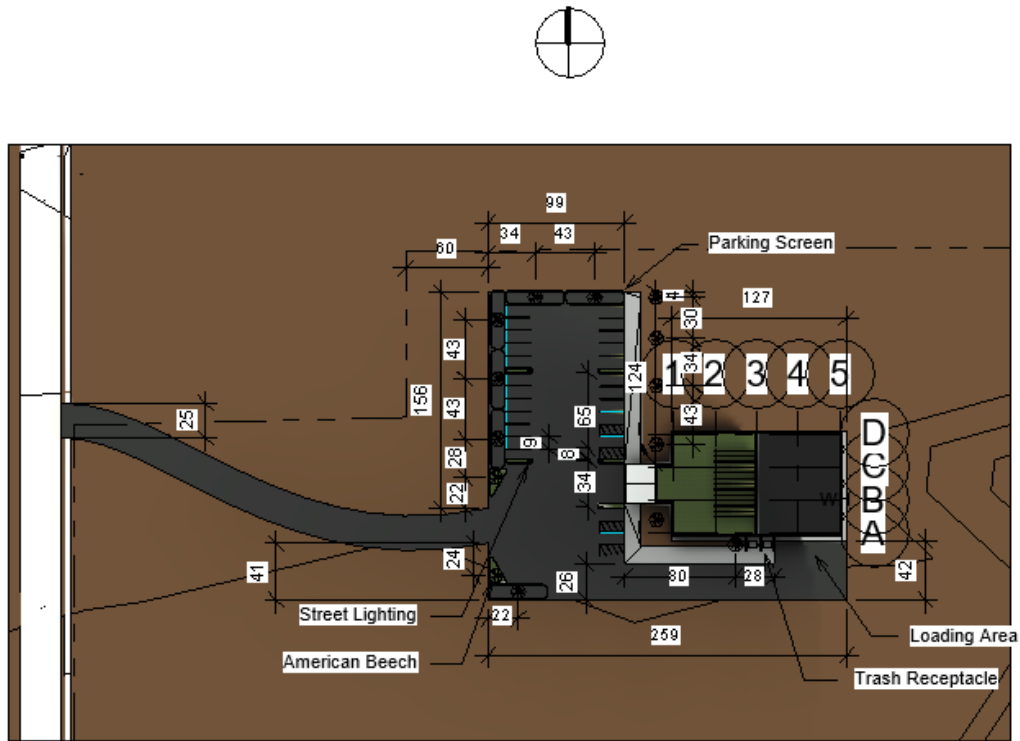
	Client	Rev.	Revision Description	Date	Sheet Size A Landscape	
	Keystone Library				Sheet No. # of #	
	Designer Connor Langsford				Date 2/25/25	Revision 1

Section View



Landscaping

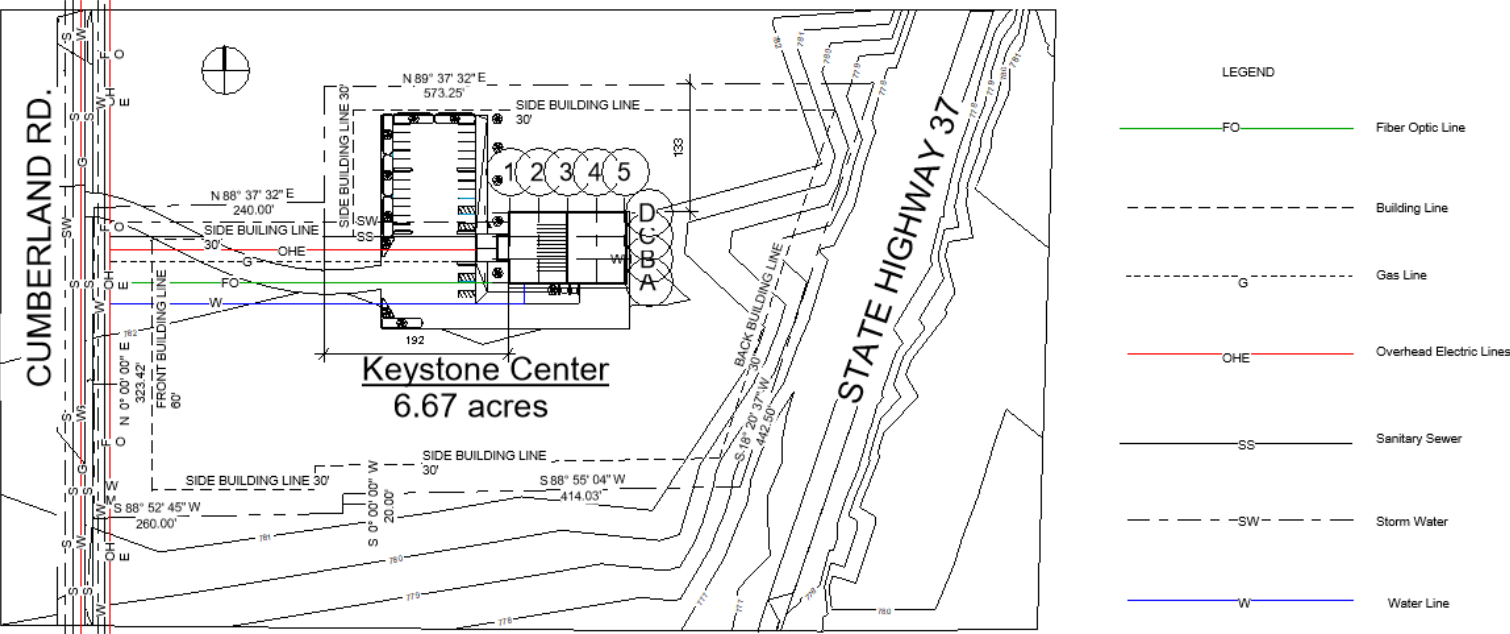
14 Evergreen 20'



① KEYSTONE CENTER SITE PLAN Copy 1
1" = 100'-0"

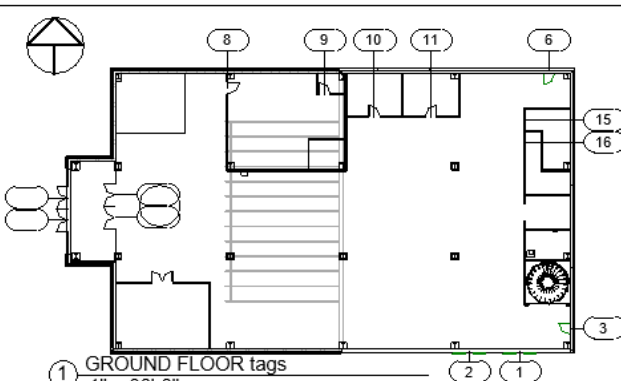
 Hamilton Southeastern	Client	Keystone Library	Rev.	Revision Description	Date	Sheet Size A Landscape	
	Designer					Sheet No.	# of #
						Date	Revision
						2/27/25	#

Site Plan

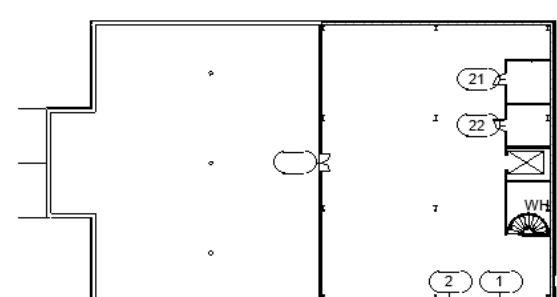


Schedules

Door Schedule		
Width	Family and Type	Mark
4' - 8 9/32"	Curtain Wall Dbl Glass: Curtain Wall Dbl Glass	
6' - 0"	Curtain Wall Dbl Glass: Curtain Wall Dbl Glass	
6' - 0"	Curtain Wall Dbl Glass: Curtain Wall Dbl Glass	
6' - 0"	Curtain Wall Dbl Glass: Curtain Wall Dbl Glass	
6' - 0"	Curtain Wall Dbl Glass: Curtain Wall Dbl Glass	
9' - 0"	Overhead-Sectional: 9 X 8	1
9' - 0"	Overhead-Sectional: 9 X 8	2
3' - 0"	Single-Flush: 36" x 84"	3
3' - 0"	Single-Flush: 36" x 84"	6
3' - 0"	Single-Flush: 36" x 80"	8
3' - 0"	Single-Flush: 36" x 80"	9
3' - 0"	Single-Glass 1: 36" x 84"	10
3' - 0"	Single-Glass 1: 36" x 84"	11
0' - 0"	Door-Opening: 36" x 84"	15
0' - 0"	Door-Opening: 36" x 84"	16
3' - 0"	Single-Flush: 36" x 80"	21
3' - 0"	Single-Flush: 36" x 80"	22
5' - 8"	Double-Glass 1: 68" x 80"	23



① GROUND FLOOR tags
1" = 30'-0"



② 2ND FLOOR Tags
1" = 30'-0"

 Hamilton Southeastern	Client	Keystone Library	Rev.	Revision Description	Date	Sheet Size	A Landscape
	Designer Connor Langsford					Sheet No.	# of #
						Date 2/25/25	Revision 1

No windows (only curtain walls)